SWINBURNE UNIVERSITY OF TECHNOLOGY

TNE20003 – Internet and Cybersecurity for Engineering Applications

Portfolio Task – Lab 5 Credit Task

Aims:

- To observe and investigate the functionality of the TCP and UDP protocols at the transport layer.
- Observe NAT at work and understand the translation process

Preparation:

- View "Transport Layer Services" & "NAT & DHCP"

Due Date:

- Your Task will be assessed via an online quiz. You must score the required minimum to pass the test. You will be allowed a number of attempts to pass the test at the grade level you attempt. You are encouraged to complete the test at the end of the lab but if you do not, you must complete it before your next lab class.

Task 1.

Build the network provided in figure 1 with Cisco Packet Tracer

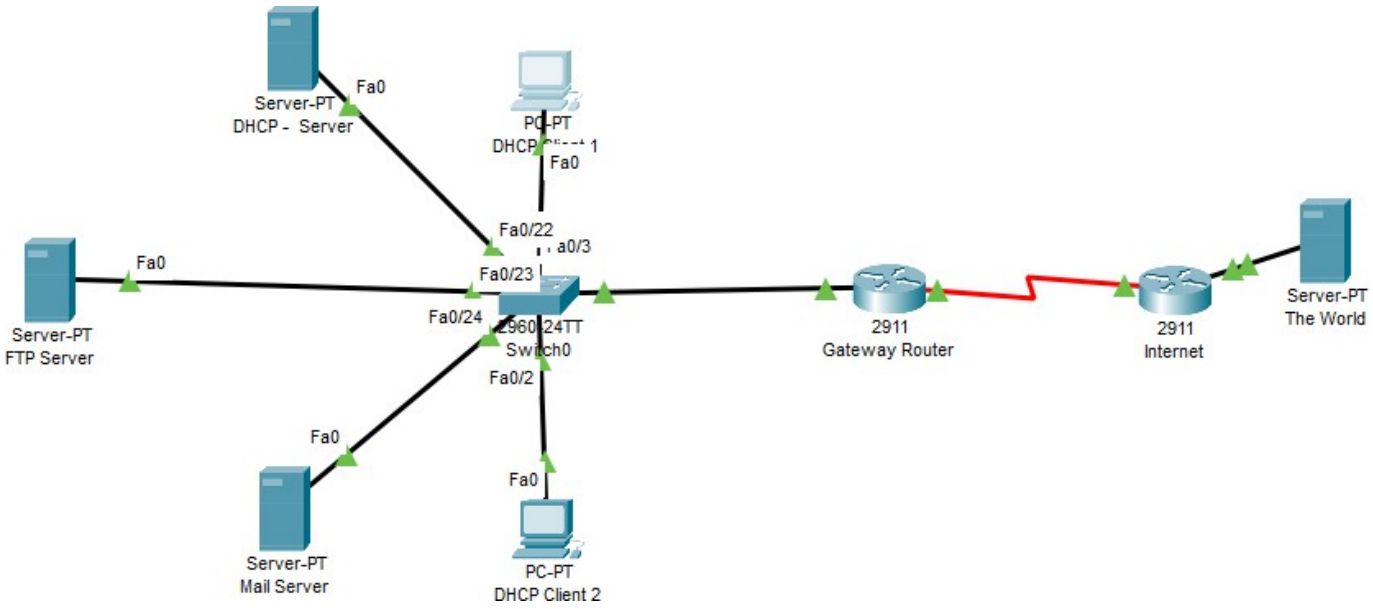


Figure 1

Use the network you built and tested in the Pass task for lab 5 to answer the following questions.

Task 2.

1. Alter the DHCP pool of addresses to 2 instead of 100.

    a. Add another DHCP Client 3 and describe what happens when this client tries to get an IP address.
    After setting the max users to 2, the third client requests an IP but it is failed because pool is full.

    b. Show this outcome on the simulation. What happens to the broadcast packets sent by the host seeking an IP address?

    The packet only reach to DHCP server.

c. What are the src and dst port numbers used? Does it make sense?

The src port 68 and dst port 67. This make sense because DHCP communication in use UDP port 68 use for client and port 67 use for server.

d. What are the src and dst IP addresses?

The src IP :0.0.0.0 and dst ip is 255.255.255.255.

e. Which transport layer protocol is being used? Explain why.

The UDP transport layer protocol is used for DHCP because it provide an efficient IP address assignment in network.

f. If you alter the DHCP pool of addresses to 3 and carry the request from client 3 again what happens? Why?

The DHCP pool will fulfill the request from Client 3 because there are three IP addresses in the pool, and only two clients have been assigned IPs, so the remaining IP address will be assigned to Client 3, which requested it.

2. Let's investigate the **FTP** server a little bit more.

a. Log into the **ftp** server with username **Dragan** and password **Fire** as in the Pass task.

b. Analyse the packet when you type "**dir**" to see what files are available for copying. What are the src and dst ports used?

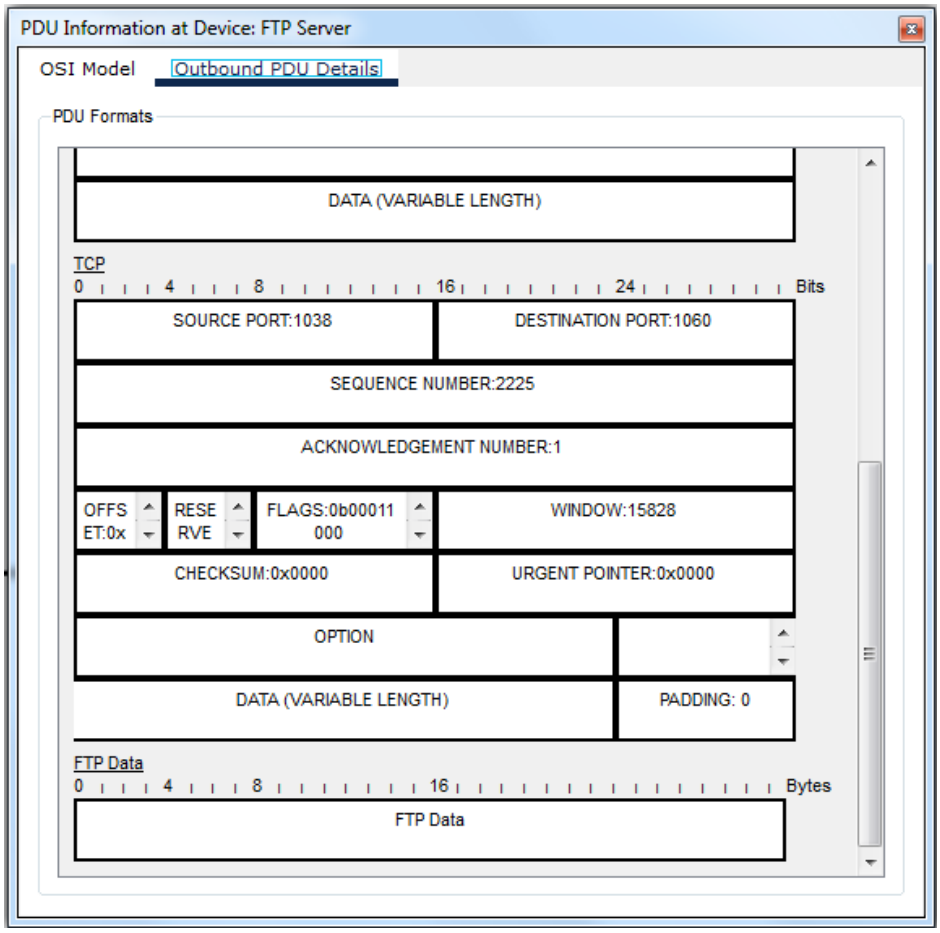
The src port 1030 and dst 21.

```
ftp>dir

Listing /ftp directory from 192.168.1.252:
0   : asa842-k8.bin                                  5571584
1   : asa923-k8.bin                                  30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin         33591768
3   : c1841-ipbase-mz.123-14.T7.bin                  13832032
4   : c1841-ipbasek9-mz.124-12.bin                   16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin         33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin         33591768
7   : c2600-i-mz.122-28.bin                          5571584
8   : c2600-ipbasek9-mz.124-8.bin                    13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin       50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin        33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                  15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin         33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin                 3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin                 3117390
16  : c2960-lanbase-mz.122-25.FX.bin                 4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin               4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin               4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin        8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin         10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin           33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin          83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin         505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG            159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG            184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin       160968869
27  : ir800-universalk9-mz.SPA.155-3.M               61750062
28  : ir800-universalk9-mz.SPA.156-3.M               63753767
29  : ir800_yocto-1.7.2.tar                          2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar             6912000
31  : pt1000-i-mz.122-28.bin                         5571584
32  : pt3000-i6q4l2-mz.121-22.EA4.bin                3117390
ftp>
```

c. Which part of the ftp process does this output represent? See picture below:

This output represents the file which have been forward from client to server.


PDU Information at Device: FTP Server
OSI Model
Outbound PDU Details
PDU Formats
DATA (VARIABLE LENGTH)
TCP
0
4
8
16
24
Bits
SOURCE PORT:1038
DESTINATION PORT:1060
SEQUENCE NUMBER:2225
ACKNOWLEDGEMENT NUMBER:1
OFFSET:0x
RESERVE
FLAGS:0b000011000
WINDOW:15828
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
FTP Data
0
4
8
16
Bytes
FTP Data

~~~~~ End of Lab ~~~~~